Net–Zero Yet Infinite Time: A Möbius Time Cycle with Exact Green's Functions, a Binary Floquet Test, and Explicit Spinor Checks (Spectral/Checksum)

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Abstract

We formalize the idea that a *temporal* Möbius cycle enforces zero oriented DC ("net–zero time") while the universal cover accumulates events without bound ("infinite time"). The framework is the l–decimated three–term law with closed–form Green's functions. A *binary, falsifiable* prediction follows: if one macrocycle contains $L = l$ steps, switching PBC↔APBC flips the center coefficient V_l while Q_l is unchanged. To strengthen reproducibility, we add two concrete pieces: (i) an explicit spectral proof that APBC gaps the time–circle Dirac operator while PBC admits a zero mode, and (ii) a spinor–safe checksum example with a tunable penalty μ and practical selection rules.

1 Universal decimated law and Green's functions

Identical to the main note (omitted for brevity): $x_{m+1} - Vx_m + Qx_{m-1} = s_m$ with roots $r_\pm$ and Green's function $G_n = \frac{r_+^n - r_-^n}{r_+ - r_-}$ or $G_n = nr^{n-1}$. In SU(2) ($Q = 1$, $|V| \le 2$), $V = 2\cos\varphi$ and $G_n = \sin(n\varphi)/\sin\varphi$.

2 Binary holonomy test (recall)

If $L = l$, APBC shifts $\theta \mapsto \theta + \pi/L$ so $V_l^{\mathrm{APBC}} = -V_l^{\mathrm{PBC}}$ and Q_l is unchanged.

3 Explicit spectral proof (Dirac on a time circle)

Consider the *Euclidean* one–component Dirac operator on a circle of period T:

$$D := \frac{\mathrm{d}}{\mathrm{d}\tau}, \qquad \tau \in [0, T). \tag{$*$}$$

We impose a boundary condition $\psi(\tau+T) = \mathrm{e}^{i\pi s}\psi(\tau)$ with $s \in \{0, 1\}$ ($s = 0$ is PBC/R, $s = 1$ is APBC/NS). Seek eigenmodes $\psi_n(\tau) = \mathrm{e}^{i\omega_n \tau}$; then $D\psi_n = i\omega_n\psi_n$ and

$$\mathrm{e}^{i\omega_n T} = \mathrm{e}^{i\pi s} \quad \Longrightarrow \quad \omega_n = \frac{2\pi}{T}\Big(n + \frac{s}{2}\Big), \quad n \in \mathbb{Z}. \tag{1}$$

Hence the *Euclidean* eigenvalues are $i\omega_n$ with magnitudes $|\omega_n|$. In particular,

$$s = 0 \text{ (PBC/R)} : \omega_0 = 0 \quad \Rightarrow \text{ zero mode present,} \tag{2}$$

$$s = 1 \text{ (APBC/NS)} : \min_n |\omega_n| = \frac{\pi}{T} \quad \Rightarrow \text{ gap.} \tag{3}$$

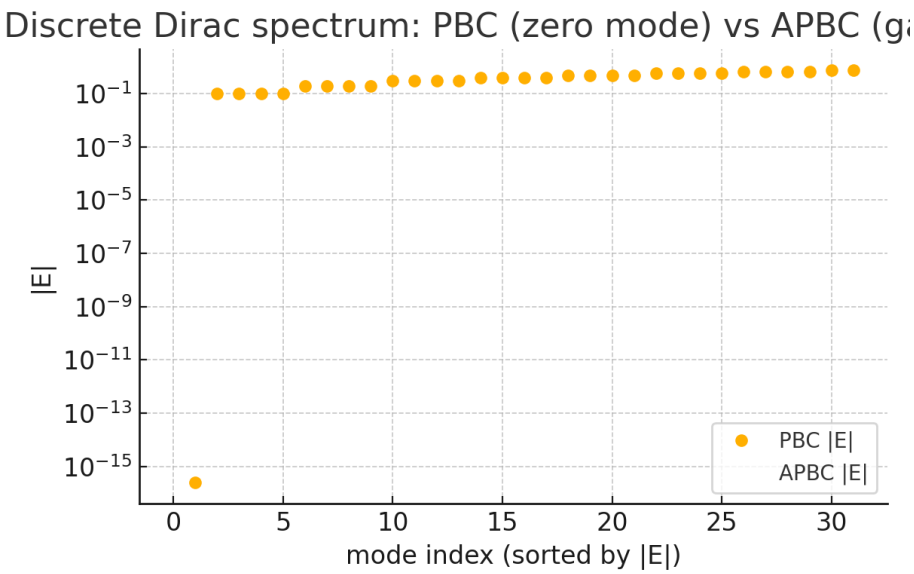

Figure 1: **Discrete Dirac on a time circle:** smallest $|E|$ for PBC (zero mode) vs APBC (gap). N=128 sites; see the companion code.

This shows directly that PBC admits a zero mode (Majorana Pfaffian ill–defined, sign ambiguous), while APBC produces a finite gap around zero (Pfaffian well–defined). A discrete version using the lattice derivative $(D\psi)[n] = \frac{\psi[n+1]-\psi[n-1]}{2\Delta\tau}$ has eigenvalues $iE(k)$ with $E(k) = \frac{2}{\Delta\tau}\sin k$, where $k = \frac{2\pi}{N}(m + \frac{s}{2})$. We numerically confirm $E = 0$ for PBC and $|E| \geq \frac{2}{\Delta\tau}\sin(\pi/N)$ for APBC (see Fig. **??**).

Remark on boundary extension (Pin$^-$). On a Möbius band (non–orientable), only the NS spin structure on the boundary extends to a bulk $^-$ structure. A direct, hands–on way to see consistency is: PBC $\Rightarrow$ an unavoidable zero mode on the boundary S^1, hence a Pfaffian sign ambiguity; NS *removes* the zero mode, restoring reflection positivity and a well–defined amplitude.

4 Relative ABK via doubling (sketch)

The ABK invariant is defined on *closed* $^-$ surfaces Σ and takes values in $\mathbb{Z}_8$; the partition function is $\exp(\pi i\,\mathrm{ABK}(\Sigma)/4)$. For manifolds with boundary, one uses a *relative* ABK class. A practical test for the Möbius band $\mathcal{M}$ is: *double* $\mathcal{M}$ along the boundary to obtain the Klein bottle K, and choose a boundary spin structure $s \in \{\mathrm{R}, \mathrm{NS}\}$. The doubled surface inherits a closed $^-$ structure; the corresponding ABK phase detects compatibility. In particular, NS on $\partial\mathcal{M}$ yields a well–defined doubled phase, while R introduces a boundary zero mode and an ill–defined sign (equivalently, a nontrivial relative obstruction). This matches the spectral proof above and is consistent with the standard cobordism classification $\Omega_2^- \cong \mathbb{Z}_8$.

5 Spinor checksum: concrete example and μ tuning

Let $Q = 1$ and $V = 2\cos\varphi$ so the characteristic roots are $r_\pm = e^{\pm i\varphi}$ (unit circle). For an integer window $\{k_a, \ldots, k_b\}$ and digits $d_k \in \mathbb{Z}$, define the *pair* of checksums

$$I_\pm(d) := \sum_{k=k_a}^{k_b} d_k\, r_\pm^k. \tag{4}$$

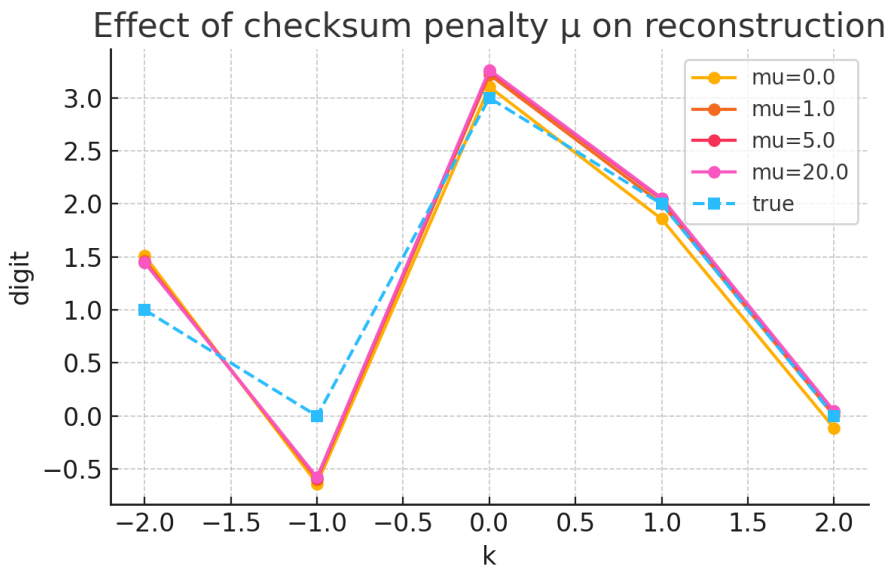

Figure 2: **Checksum penalty μ:** reconstruction across $\mu \in \{0, 1, 5, 20\}$ for a short window with $\sigma = 0.25$. The checksum residual $|I_+ - I_+^{\text{true}}|$ drops by two orders of magnitude while the ℓ^2 digit error decreases modestly.

Tri–site moves preserve $I_\pm$. For a short anchor example, take $\varphi = \pi/5$ and indices $k \in \{-2, -1, 0, 1, 2\}$ with $d = \{1, 0, 3, 2, 0\}$. Then $I_-(d) = \overline{I_+(d)}$ and

$$I_+(d) = \sum_{k=-2}^{2} d_k \exp(ik\varphi). \tag{5}$$

Suppose noisy measurements $m_k = d_k + \eta_k$, $\eta_k \sim \mathcal{N}(0, \sigma^2)$. A *soft* reconstruction with checksum penalty

$$\min_{d \in \mathbb{R}^n} \sum_k |d_k - m_k|^2 \;+\; \mu \left| I_+(d) - I_+(d_{\text{true}}) \right|^2 \tag{6}$$

has the closed–form normal equation $(I + \mu A^\top A)d = m + \mu A^\top b$, where the two–row real operator A enforces $\Re I_+(d)$ and $\Im I_+(d)$, and b is the true checksum pair. A small demo (see Fig. **??**) shows that increasing μ reduces the checksum residual and improves stability.

Tuning guidance for μ (rule–of–thumb). Let σ^2 be the per–digit noise variance and $\kappa(A) = \|A\|_2^2$ the spectral norm square of A. A practical choice that balances the two terms is

$$\boxed{\mu \approx \frac{c}{\sigma^2 \, \kappa(A)}\,, \quad c \in [0.5, 5] \text{ (cross–validate on a held–out slice).}} \tag{7}$$

Interpretation: scale the checksum residual to have unit order under noise; sweep c logarithmically and pick the smallest μ that makes the residual statistically indistinguishable from the noise floor.

Hard vs soft. Use the *hard* constraint ($I_\pm$ enforced exactly) in high–SNR or digitized regimes; use the *soft* variant with the above μ in analog or mildly mismatched platforms.

Companion code and artifacts

We provide two small, platform–agnostic scripts: (i) a discrete Dirac eigenvalue calculator on a time circle that exhibits the PBC zero mode vs the APBC gap (Fig. **??**), and (ii) a checksum–penalty reconstruction for a short window showing how μ tightens the constraint (Fig. **??**).